DOACs in Patients with Obesity

Objective:

To provide guidance about the management of direct oral anticoagulants (DOACs) in patients with obesity.

Background:

Four DOACs are approved for clinical use in Canada: apixaban, dabigatran, edoxaban and rivaroxaban. The clinical guides for each of the DOACs contains information about mechanism of action, licensed indications, side-effect profiles, and approved dosing regimens [see **Clinical Guides Apixaban (Eliquis®), Dabigatran (Pradaxa®), Edoxaban (Lixiana®), and Rivaroxaban (Xarelto®)]**. According to Health Canada, obesity is defined as a body mass index (BMI) >30 kg/m^2.

Pharmacokinetics of DOACs in obese patients

Obesity is known to have variable effects on pharmacokinetic parameters (absorption, distribution, metabolism, elimination), which could influence the overall exposure of these anticoagulants. The product monographs for three of the DOACs (apixaban, dabigatran, rivaroxaban) report the results of pharmacokinetic analyses demonstrating that weights >120kg are associated with 20-30% decrease in overall drug exposure. This difference is not considered to be clinically significant.

Clinical efficacy and safety of DOACs in obese patients:

No randomized controlled trials have examined the safety and efficacy of DOACs specifically in patients with obesity. In most instances, <20% of patients enrolled in DOAC trials weighed >90-100 kg or had a BMI >30 kg/m^2. Published subgroup analyses of these trials have consistently shown that DOACs have a similar efficacy and safety profile when compared with warfarin in obese populations. A high weight alone does not necessarily indicate obesity, thereby limiting the interpretation of data from trials that reported weight only.

- **Studies in venous thromboembolism:** In a systematic review and meta-analysis of patients with VTE treated with apixaban, dabigatran, and rivaroxaban, anticoagulant efficacy was similar in patients <100 kg and ≥100 kg. Another study found no association between BMI and risk of VTE recurrence or bleeding events in rivaroxaban-treated patients. A third retrospective study of apixaban vs warfarin used pooled data from various U.S. databases to analyze outcomes in over 43,000 patients with VTE who were classified as either obese or morbidly obese. No significant difference in recurrent VTE or major bleeding were identified (slightly lower risk for both outcomes in the apixaban arm). Subgroup analysis of patients in the Hokusai VTE study with edoxaban showed no significant difference in efficacy or safety in the >100kg or <100kg groups. An observational study using propensity matched cohorts found benefit for rivaroxaban compared to warfarin in reducing VTE recurrence in patients with BMI ≥35 kg/m^2 and reduced risk of major bleeding.
- **Studies in atrial fibrillation:** A systematic review and subsequently published post-hoc subgroup analyses of large, randomized trials suggest that, among atrial fibrillation patients, apixaban, dabigatran, rivaroxaban and edoxaban are equivalent to vitamin K antagonists in terms of efficacy and safety in patients

retrospective study was done at an integrated multisite health-care system of DOAC use a
categories. No differences were found in stroke rates or risk of intracranial hemorrhage bet
and those with category 1 or higher obesity (43.9% apixaban, 36.4% rivaroxaban, 9.5% da
edoxaban).

- **Studies in orthopedic surgery:** In a study that compared clinical outcomes in patients undergoing total hip and knee arthroplasty, no significant differences were noted between the efficacy and safety of dabigatran and enoxaparin in patients with a BMI >30 kg/m^2. A study evaluating patients undergoing total hip or knee arthroplasty found no differences in the reduction in venous thromboembolism (VTE) risk with rivaroxaban versus enoxaparin in patients >90kg compared with those with lower weights. Finally, in a subgroup analysis of pooled data from studies comparing apixaban and enoxaparin, there were no differences in VTE or bleeding events when patients with BMI<30 kg/m^2 were compared with those with a BMI $\geq$30 kg/m^2.

Use of DOACS in patients with Class III obesity (BMI $\geq$ 40 kg/m^2):

Data and clinical experience supporting the use of DOACs in patients with a BMI >40 kg/m^2 or weight >120 kg remain limited to retrospective and observational studies, but clinical efficacy and safety results are encouraging. Randomized controlled trial testing of DOACs vs. warfarin in this population are needed to validate these findings.

- Three separate retrospective studies of patients with BMI ≥40 kg/m^2 found similar rates of recurrent VTE and major bleeding when comparing rivaroxaban and warfarin, one of which found similar results with apixaban versus warfarin.
- A systematic review of 5 observational studies comprising real-world data has been carried out on patients with body weight >120kg or BMI >40 kg/m^2 and acute VTE (n=6585). The major DOACs used were rivaroxaban and apixaban. The meta-analysis concluded that these DOACs are non-inferior to warfarin in reducing the primary efficacy outcome of VTE recurrence (OR 1.07, 95% CI 0.93-1.23) and primary safety outcome of major bleeding (OR 0.80, 95% CI 0.54-1.17).
- A systematic review and meta-analysis of 10, mostly retrospective cohort studies, comprising 89,494 very obese (BMI >40 kg/m^2 or weight >120 kg) patients with non-valvular atrial fibrillation on oral anticoagulation therapy (45,427 on DOACs vs. 44,067 on warfarin) was conducted in 2021. The rate of stroke and systemic embolism was significantly lower in DOAC group compared with warfarin group (OR 0.71, 95% CI 0.62–0.81, P <0.0001). The major bleeding rate was also significantly lower in the DOAC group compared with the warfarin group (OR 0.60, 95% CI 0.46–0.78, P <0.0001).
- A post-hoc analysis of the ARISTOTLE randomized controlled trial (apixaban vs warfarin in atrial fibrillation) compared outcomes stratified by weight [among 18,139 patients, 10.5% were <60 kg, 83.6% were 60-120 kg, and 5.4% were >20 kg]. Across all weight categories, efficacy (prevention of stroke or systemic embolization) was similar for both anticoagulants while major bleeding was lower in the apixaban arm.
- In a post-hoc analysis of the ENGAGE AF-TIMI 48 randomized controlled trial (edoxaban vs warfarin in atrial fibrillation), the efficacy and safety profile of edoxaban were similar across all BMI categories [among 21,028 patients, 0.8% had a BMI <18.5 kg/m^2, 21.4% had a BMI of 18.5-25 kg/m^2, 37.6% had a BMI of 25-30 kg/m^2, 24.8% had a BMI of 30-35 kg/m^2, 10.0% had a BMI of 35-40 kg/m^2, and 5.5% had a BMI ≥40 kg/m^2].

DOAC drug level monitoring:

There is currently insufficient clinical data to recommend routine monitoring of DOAC levels in patients with obesity to assess for efficacy. This is primarily due to the fact that there are no accepted evidence-based therapeutic

targets available. In addition, there can be significant inter- and intra-patient variability when in
Moreover, availability of DOAC drug level testing is significantly limited.

Recommendations:

- In patients weighing up to 120 kg or with a BMI up to 40 kg/m2, there is sufficient evidence available demonstrating that DOACs can be used safely in this patient population.
- For patients weighing >120 kg or with a BMI >40 kg/m^2, the ISTH recommends standard dosing rivaroxaban or apixaban as appropriate options for treatment of VTE, with less data supporting apixaban than rivaroxaban. The use of dabigatran or edoxaban is not recommended given unconvincing data for the former and lack of pharmacokinetic/pharmacodynamic data for the latter. It should be noted that there is no widely accepted weight or BMI limit above which DOACs should not be considered. With increasing weights and BMIs, there is more limited evidence for their efficacy and safety. In patients with extremely elevated weights and BMIs, decisions to use DOACs should be evaluated on a case-by-case basis, taking into account the individual thrombotic risk. Patients should be informed of the limitations of the available evidence and the potential risk of underdosing.

Special Considerations:

Data regarding safety and efficacy of DOACs in the pediatric population are very limited. DOACs are contraindicated in pregnancy and while breastfeeding.

Other Relevant

- Apixaban (Eliquis®)
- Dabigatran (Pradaxa®)
- Deep Vein Thrombosis (DVT): Treatment
- Edoxaban (Lixiana®)
- Pediatric Thrombosis
- Pregnancy: Thromboprophylaxis
- Pregnancy: Venous Thromboembolism Treatment
- Pulmonary Embolism (PE): Treatment
- Rivaroxaban (Xarelto®)
- Thromboprophylaxis: Orthopedic Surgery

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